

RYA SANOVAR

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RESEARCH INTERESTS

My research focuses on enabling hardware-efficient AI inference through the mediums of algorithms, systems and computer architecture. Currently, I am focusing on improving interactivity of agentic decode by reducing collective latencies in scale-up networks.

EDUCATION

Georgia Institute of Technology

Aug 2025 - Present

Ph.D. in Computer Science

Advised by Prof. Moinuddin Qureshi, Future Architectures and Systems Lab

Birla Institute of Technology and Science, Pilani

Oct 2020 - Jun 2024

B.E. in Electronics and Communication

GPA: 8.91/10.0

WORK EXPERIENCE

Microsoft Research

Cambridge, UK

Research Intern

May 2026 - Aug 2026

- Investigated fault tolerance for disaggregated decode: Impact of network link failures and device failures on decode system throughput and tail latencies.
- Develop cost-effective scale-out network topologies for disaggregated inference resilient to network failures at datacenter scale.

Microsoft Research

Bangalore, India

Research Fellow

Jan 2024 - Jul 2025

- Developed **LeanAttention**: A hardware-aware scalable exact-attention execution mechanism that accelerates attention by 2.6x over FlashAttention-2. Integrated into **ONNXRuntime**.
- Published and filed patents in key novel techniques to improve efficiency of LLM inference.

PUBLICATIONS

Rya Sanovar, Srikant Bharadwaj, Hritvik Taneja, Moinuddin Qureshi. “*SIFT: Selective-Index For Fast Compute of RAG Prefill by Exploiting Attention Invariance*”
Under Submission.

Rya Sanovar, Srikant Bharadwaj, Renee St. Amant, Victor Rühle, Saravan Rajmohan. “*Lean Attention: Hardware-Aware Scalable Attention Mechanism for the Decode-Phase of Transformers*”
The Eighth Annual Conference on Machine Learning and Systems (MLSys), Santa Clara, May 2025

Rya Sanovar. “*Horses for Courses: Unique Hardware Efficiency Challenges and Solutions for Prefill and Decode Inference.*”

Poster presented at the *Young Professionals Symposium @ MLSys, Santa Clara, May 2025.*

PATENTS

Rya Sanovar, Srikant Bharadwaj, Victor Rühle. “*Hardware-aware attention mechanism with dynamic workload distribution for transformer models.*”

AWARDS

ISCA Student Travel Grant

2026

RELEVANT COURSEWORK

Computer Science: Advanced Computer Architecture, Operating Systems, Machine Learning

Electrical and Electronics: Microelectronic Circuit Design, Microprocessors and Interfacing